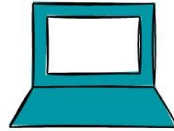


Quarkus demo - architectures outline :

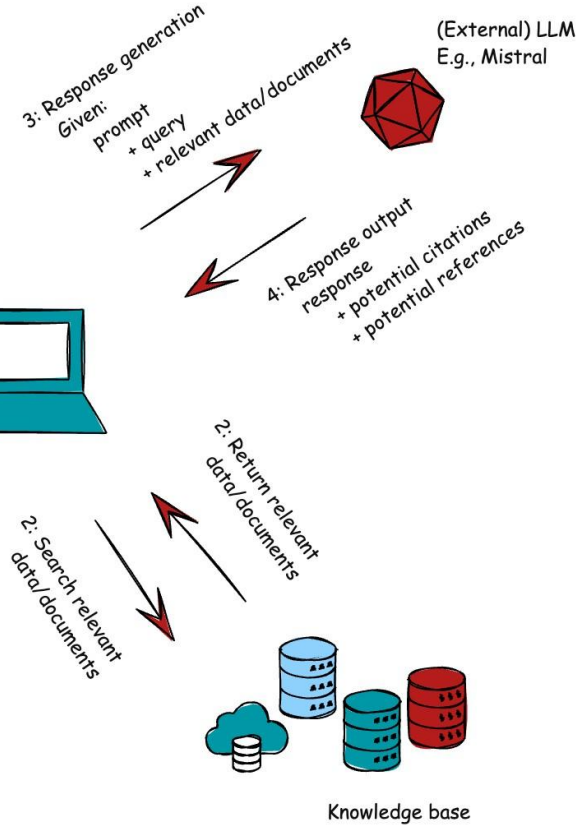
Model enhancements types/architectures



1: Query

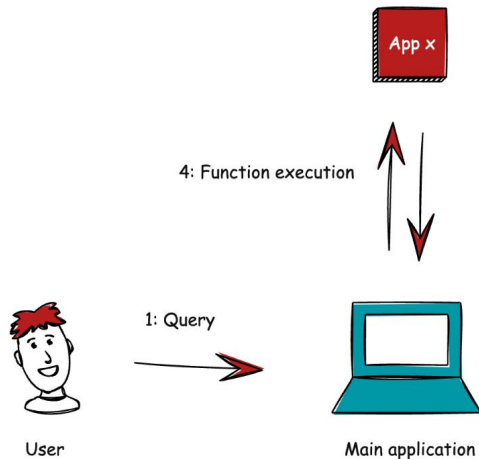


*"Why is Camille's cat,
Mitsou, the cutest cat in
the whole world?"*



RAG

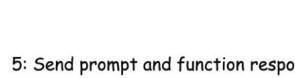
- https://redhat-developer-demos.github.io/quarkus-tutorial/quarkus-tutorial/17_prompts.html
- Protect data.
- No knowledge, use case, time, ... to fine-tune, train a model yourself.
- Source references.
- Less scalable, bigger models.
- Quarkus: next-gen java



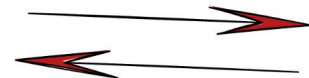
2: Send prompt and available functions



3: If response requires function call,
the LLM responds with function definition



5: Send prompt and function response



6: Final response is returned

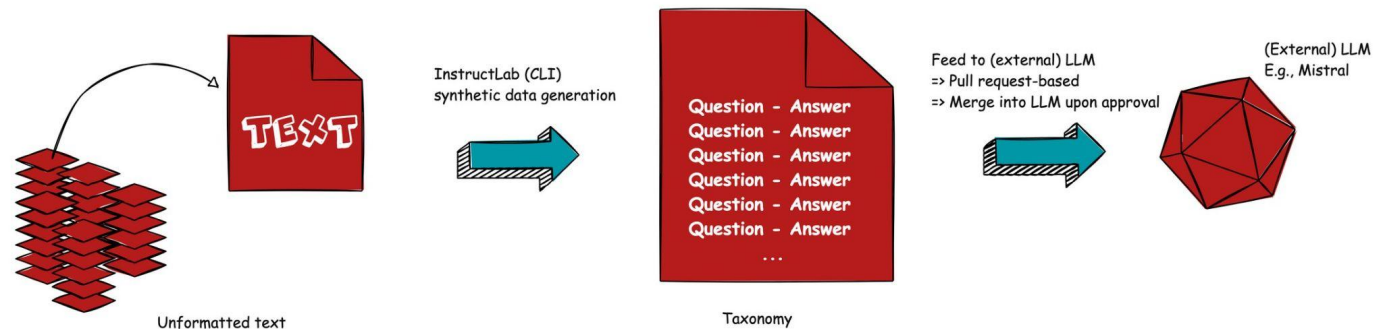
(External) LLM
E.g., Mistral



Function calling

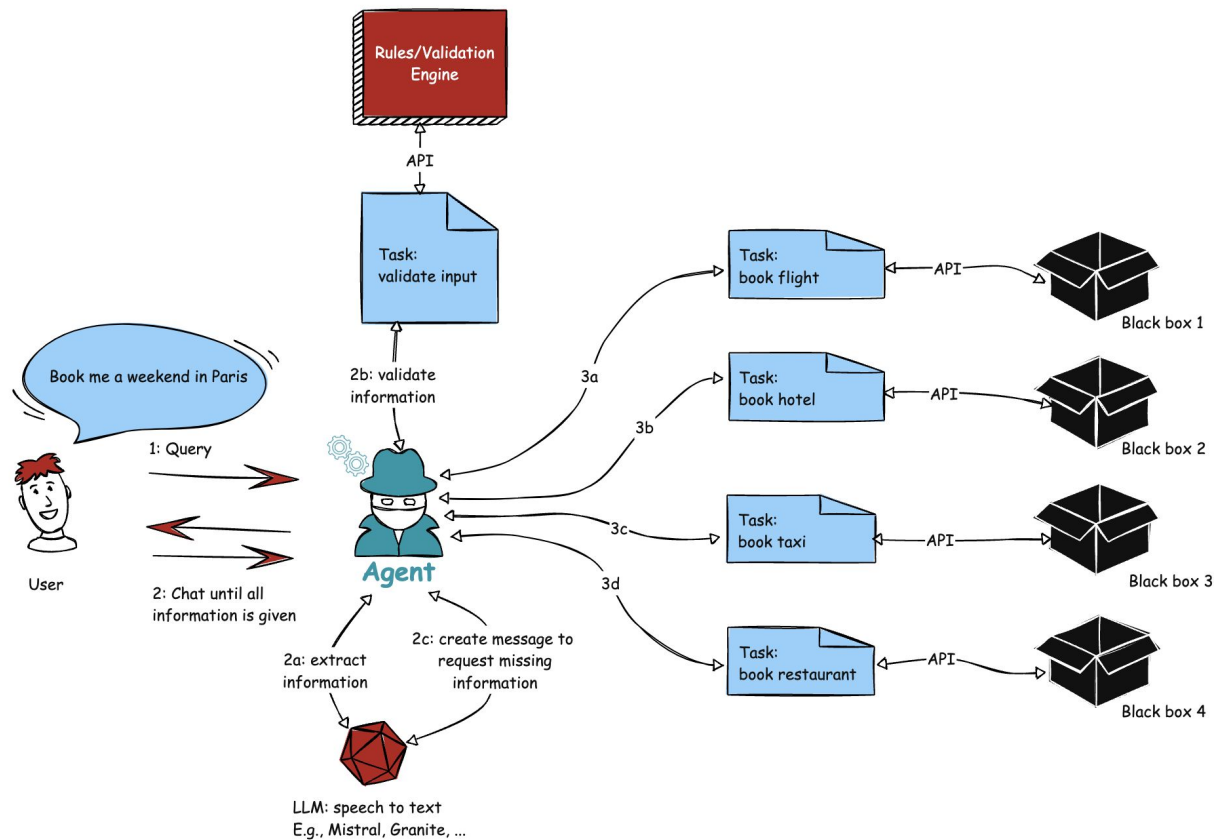
- https://redhat-developer-demos.github.io/quarkus-tutorial/quarkus-tutorial/17_prompts.html
- External call(-out).
- External APIs.
- Get factual data.

What is the weather today?



InstructLab

- <https://github.com/instructlab>
- Open collaboration.
- Open way of contributing.
- Less data science knowledge.



Agentic AI

- Quarkus.
- Camel.
- Kafka.
- ...
- Intern in prod!!



Embedding AI

Don't burn your fingers in the heat of
the AI revolution





Embedding AI

Staying cool behind the bar of the AI
revolution



**“Keep your
options open!”**



Maarten Vandeperre

maarten-vandeperre.be



Behind the bar of AI

What's really being served?



Real world news

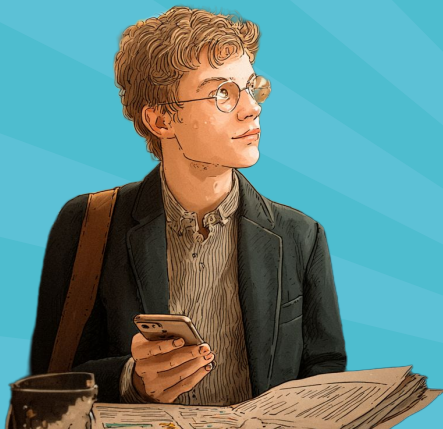
Cyber Security

What the CLOUD Act Really Means for EU Data Sovereignty

The CLOUD Act allows U.S. authorities to access data stored in the EU, putting it in direct conflict with GDPR. Learn how this impacts data sovereignty and what EU businesses can do to stay compliant

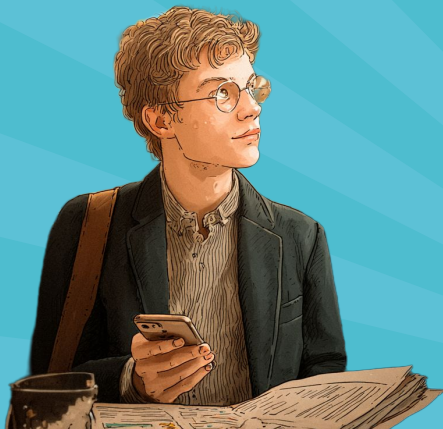


Wire
Jul 15, 2025



Source: <https://wire.com/en/blog/cloud-act-eu-data-sovereignty>

Real world news



[Redacted] EU data not protected from US access

21 July, 2025 11:35

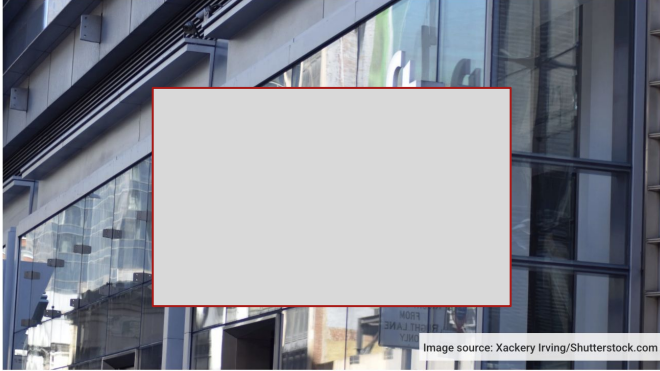


Image source: Xackery Irving/Shutterstock.com

[Facebook](#) [X](#) [LinkedIn](#) [Reddit](#) [WhatsApp](#) [Pocket](#)

 This article is also available in German.

Explosive testimony before the French Senate: [Redacted] cannot guarantee that European user data is safe from US authorities despite EU hosting.

[Redacted] cannot protect European user data from access by US authorities – even if it is stored in EU data centers. This was admitted under oath by the legal director of [Redacted] France, [Redacted], at a [hearing before the French Senate](#) on June 10, 2025.

Source: <https://www.it-daily.net/en/shortnews-en/microsoft-eu-data-not-protected-from-us-access>

Real world news

Daryna Antoniuk

June 13th, 2025

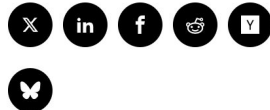
Technology

Government

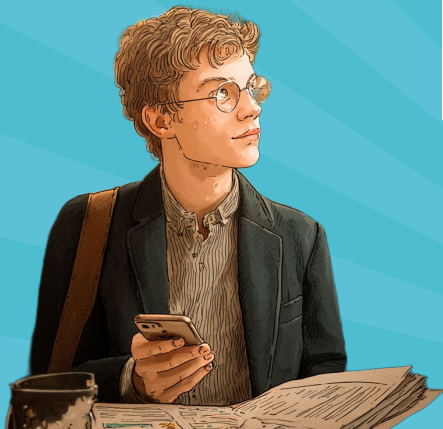
News

Danish government agency to ditch Microsoft software in push for digital independence

Denmark's tech modernization agency plans to replace Microsoft products with open-source software to reduce dependence on U.S. tech firms.



In an interview with the local newspaper *Politiken*, Danish Minister for Digitalisation Caroline Stage Olsen confirmed that over half of the ministry's staff will switch from Microsoft Office to LibreOffice next month, with a full transition to open-source software by the end of the year.



Source: <https://therecord.media/denmark-digital-agency-microsoft-digital-independence>

Real world news

NEWSLETTERS · CFO DAILY

MIT report: 95% of generative AI pilots at companies are failing



BY SHERYL ESTRADA

SENIOR WRITER AND AUTHOR OF CFO DAILY

August 18, 2025 at 6:54 AM EDT



GETTY IMAGES

Forbes

EDITORS' PICK | BUSINESS > MANUFACTURING

Why 95% Of AI Pilots Fail, And What Business Leaders Should Do Instead

By [Andrea Hill](#), Contributor. © Andrea Hill is a multi-industry CEO covering bu...

Follow Author

Published Aug 21, 2025, 11:54am EDT, Updated Aug 28, 2025, 07:54am EDT

Source: <https://www.forbes.com/sites/andreahill/2025/08/21/why-95-of-ai-pilots-fail-and-what-business-leaders-should-do-instead/>
<https://fortune.com/2025/08/18/mit-report-95-percent-generative-ai-pilots-at-companies-failing-cfo/>





Before we start pouring

Explaining key concepts



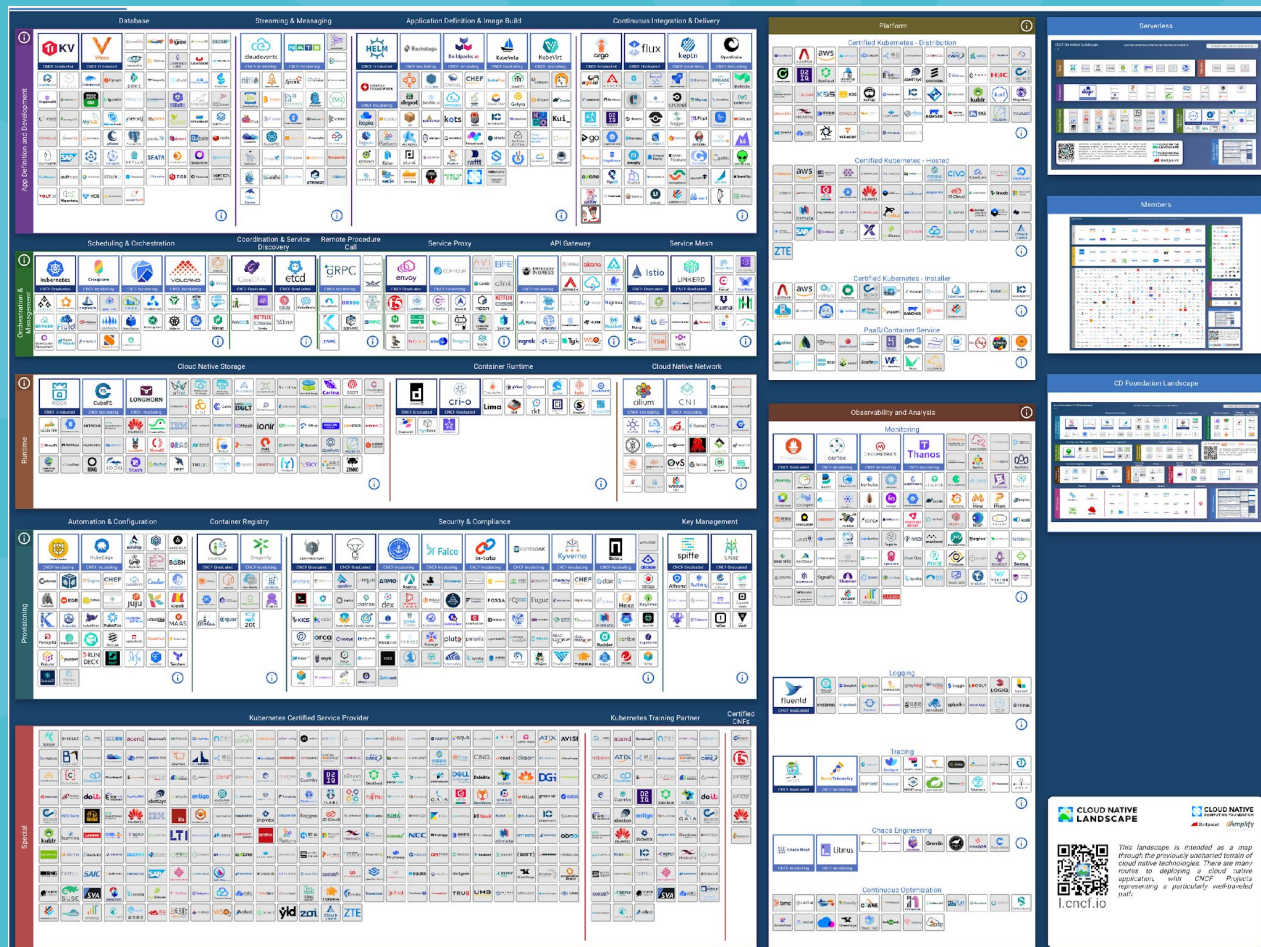
Cognitive load

The Cloud-Native Grocery Store

Where is

- Wally?
- Waldo?
- Charly?

Source: <https://landscape.cncf.io/>



Cognitive load

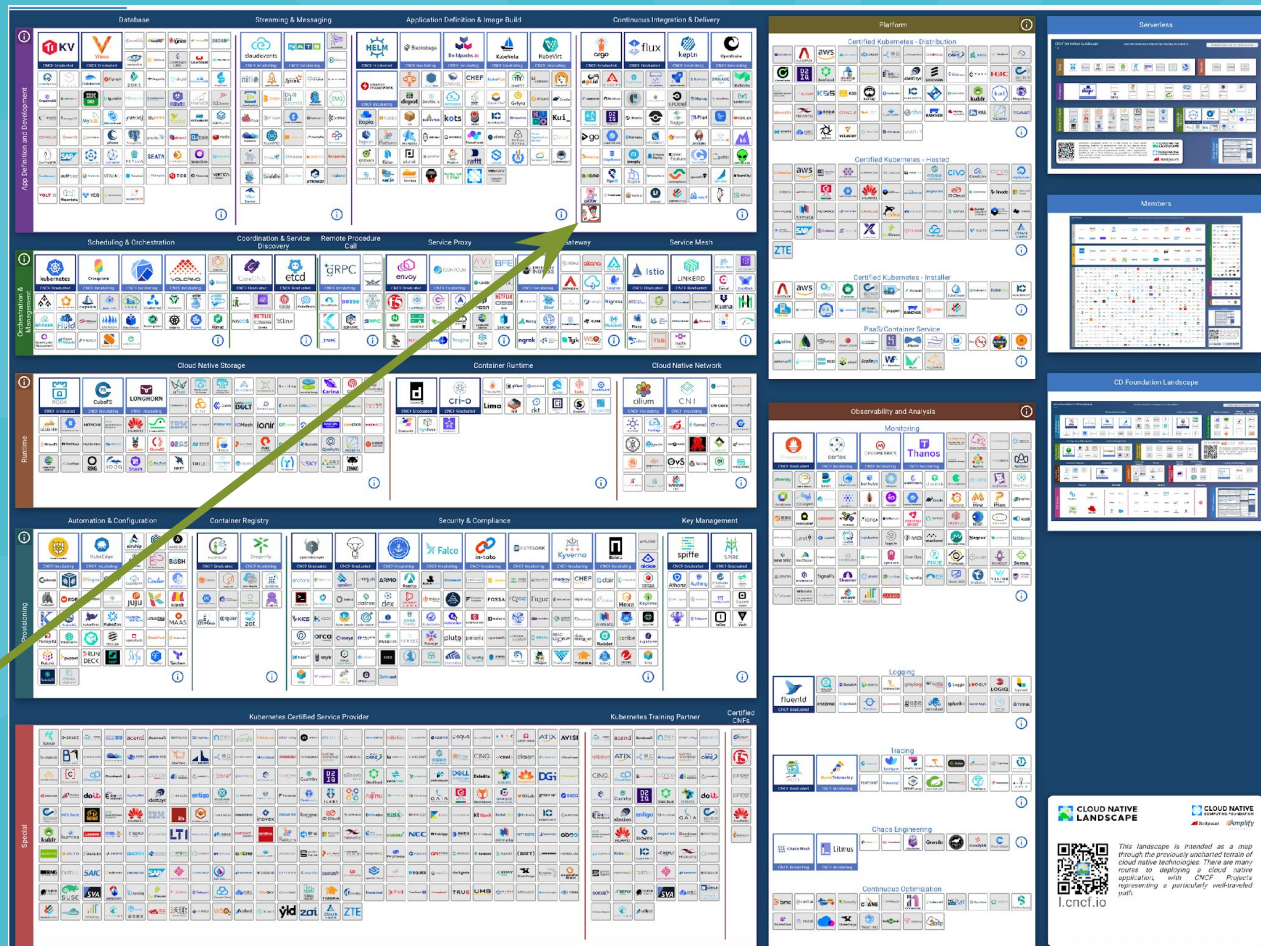
The Cloud-Native Grocery Store

Where is

- Wally?
- Waldo?
- Charly?



Source: <https://landscape.cncf.io/>



Platform engineering

Traditional App Delivery



Dev



Ops

Key:

Dev Responsible

Ops Responsible

Complete platform engineering presentation: <https://youtube.com/watch?v=DaYNwrvJGmk&si=-IDX7xiBvBixEHPP>

Platform engineering

Traditional App Delivery



Dev

Applications
App Deploy
App Build
Entitlement/Billing
Runtime
Java
Guest OS
Guest Virtualization
Clustering/HA
Host Virtualization
OS
Hardware



Ops



Key:

Dev Responsible

Ops Responsible

Modern App Delivery



Dev

Applications
App Deploy
App Build
Entitlement/Billing
Runtime
Java
Guest OS
Guest Virtualization
Clustering/HA
Host Virtualization
OS
Hardware



Ops

Usability

Observability

Quality

Platform engineering



Dev

Applications



Platform Engineering

App Build

App Deploy

Entitlement/Billing

Runtime

Java

Guest OS



Ops

Entitlement/Billing

Guest Virtualization

Clustering/HA

Host Virtualization

OS

Hardware

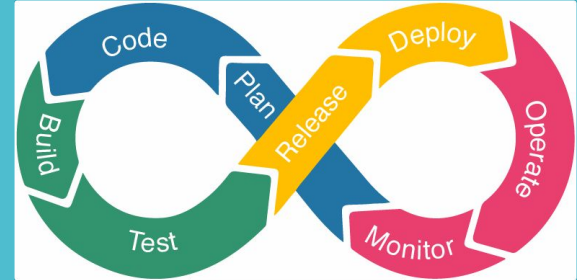
Internal Developer
Platform
IDP

!!! Product

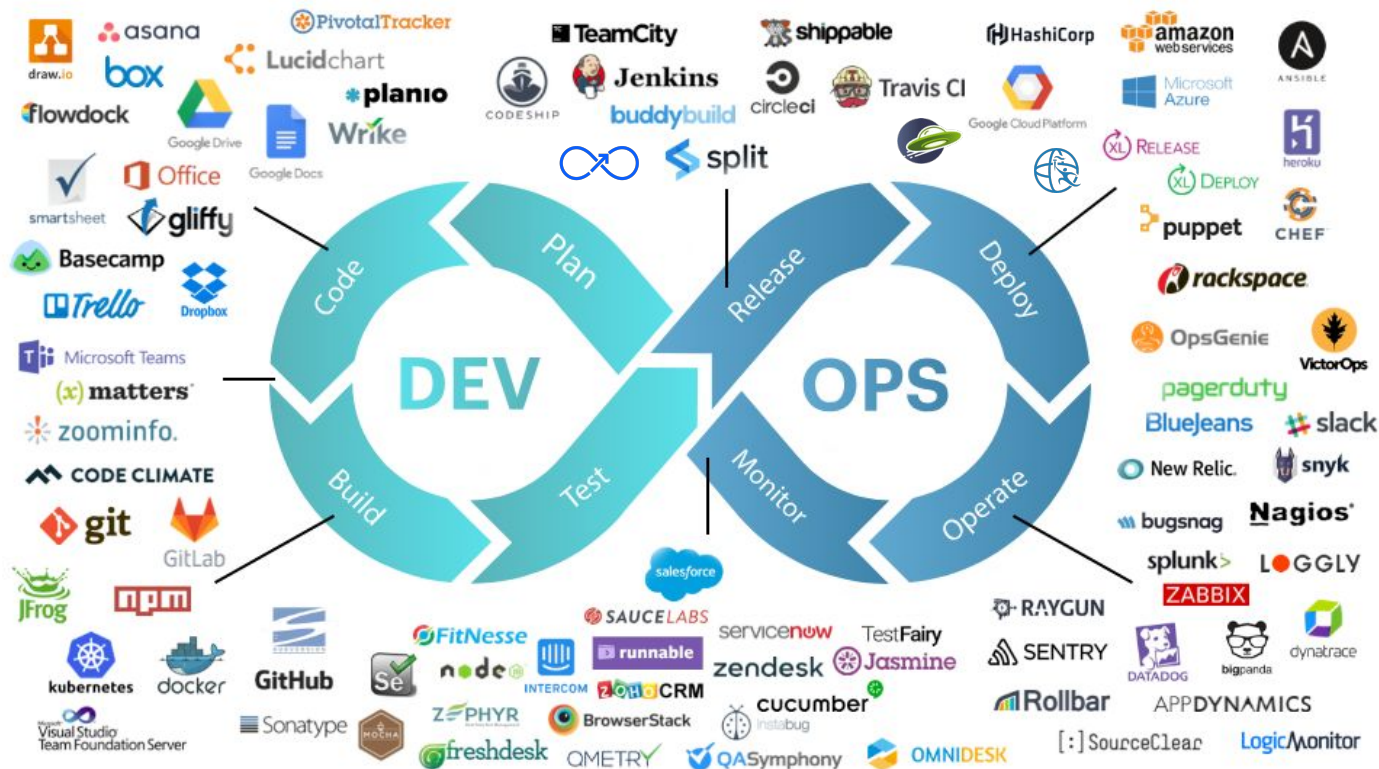
The platform engineering recipe

Ingredients

- Integrated Development Environments
- Version Control Systems
- Resources - containers, databases, storage, compute, transit, etc.
- Runtimes and frameworks
- Artefact, container registry
- Kubernetes
- Observability
- Vulnerability scanners
- CI tools
- GitOps/CD tools
- Infrastructure as Code tools

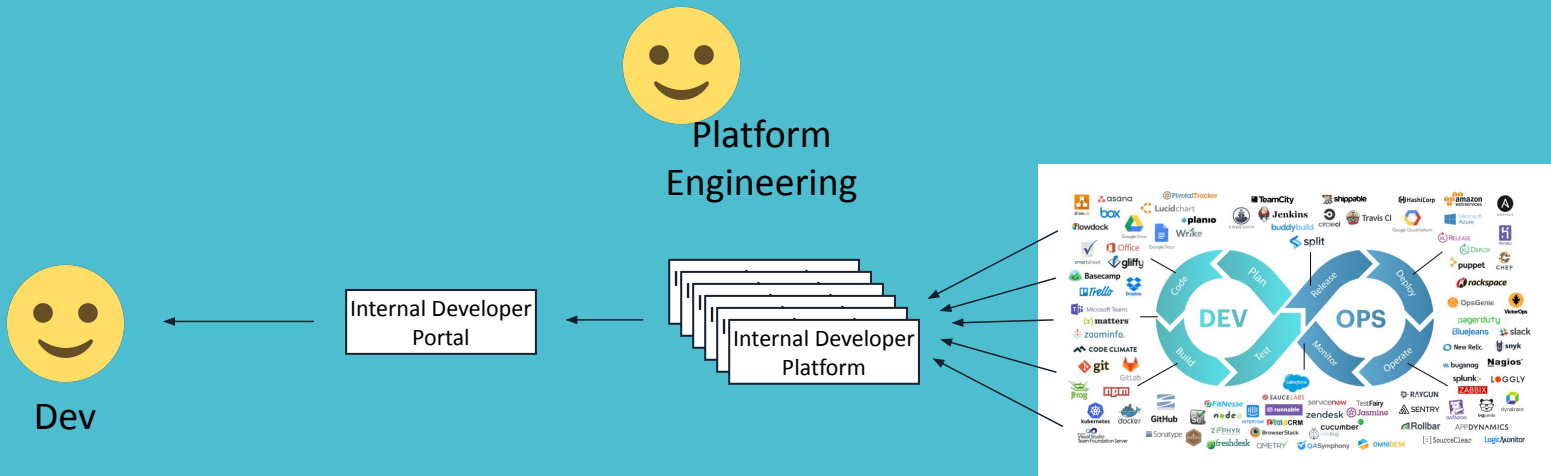


Bake your own platform



Source: <https://www.openxcell.com/devops/>

So, what does it look like?



DEMO: let's configure an AI platform (manually)

URL:

<https://developers.redhat.com>

<https://sandbox.redhat.com>

GitOps

Name	Last commit	Last update
..		
 game-secure-argocd-app-dev-build.yaml	initial commit	41 minutes ago
 game-secure-argocd-app-dev.yaml	initial commit	41 minutes ago
 game-secure-argocd-app-preprod.yaml	initial commit	41 minutes ago
 game-secure-argocd-app-prod.yaml	initial commit	41 minutes ago
 game-secure-argocd-repo.yaml	initial commit	41 minutes ago

game-secure-argocd-app-dev.yaml 1.22 KIB

Blame Edit Replace

```
1 ---
2 apiVersion: argoproj.io/v1alpha1
3 kind: Application
4 metadata:
5   name: game-secure-dev
6   namespace: janus-argocd
7   finalizers:
8     - resources-finalizer.argocd.argoproj.io
9 spec:
10   project: janus
11   source:
12     repoURL: https://gitlab-gitlab.apps.cluster-kgv66.kgv66.sandbox3262.opentlc.com/development/game-secure-gitops.git
13     targetRevision: HEAD
14     path: ./helm/app
15     helm:
16       parameters:
17         - name: namespace.name
18           value: game-secure-dev
19         - name: environment
20           value: dev
21         - name: image.tag
22           value: latest
23     destination:
24       server: https://kubernetes.default.svc
25       namespace: game-secure-dev
26   ignoreDifferences:
27     - group: '*'
28       jqPathExpressions:
29         - '.imagePullSecrets[] | select(.name | contains("-dockercfg-"))'
30     Kind: ServiceAccount
31   syncPolicy:
32     automated:
```

... Environment definitions
are stored and managed
in version control.
(and then applied)



Recap

- Terminology
 - Platform engineering
 - Cognitive load
 - GitOps
- Security & resilience challenges
- Challenges with uncontrolled (SaaS?) environments
- It is not straightforward to enable AI in production
 - ⇒ Stop "playing" with AI, make it enterprise-grade



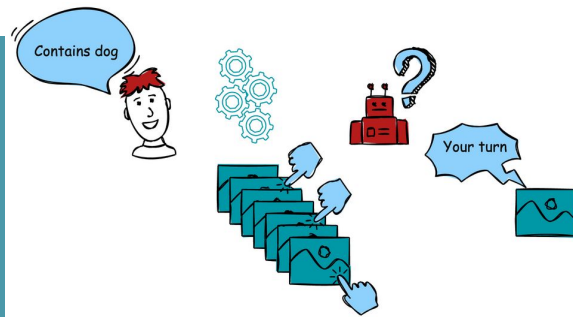
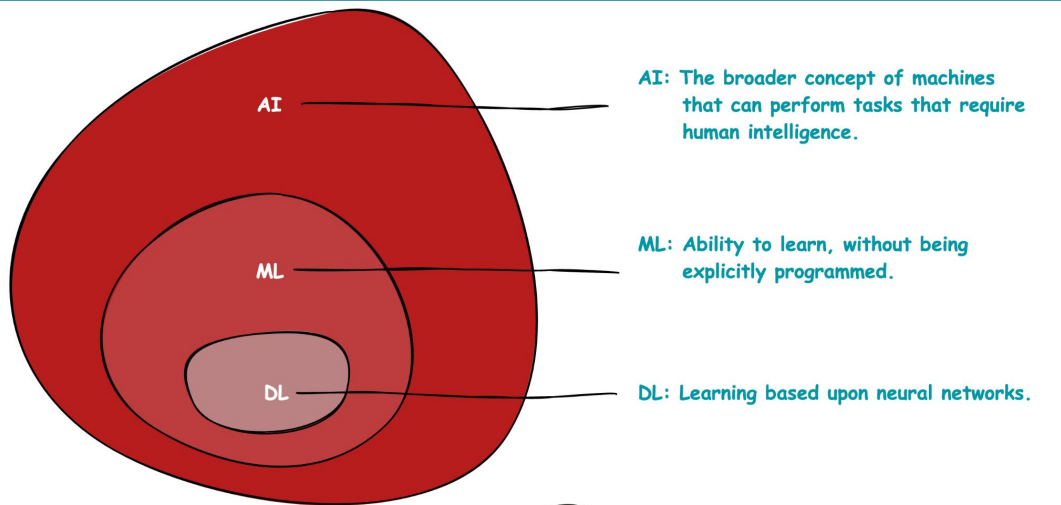


Challenges

Challenges

Which AI architecture?

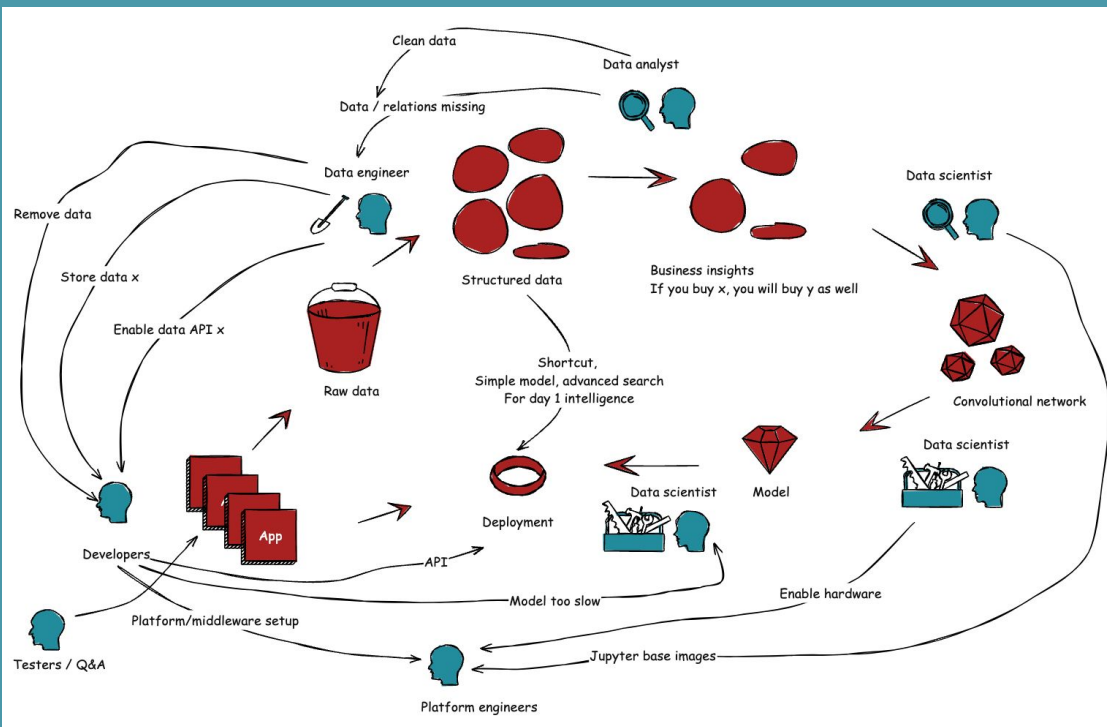




Blog:

<https://developers.redhat.com/articles/2025/04/10/road-ai-guide-understanding-ai/ml-models>

<https://youtu.be/-TG3gbSeVGO>



Challenges

Which LLM-based
architecture?



Three Friends going to a bar..



Fine-Tuna



Agent Gaga

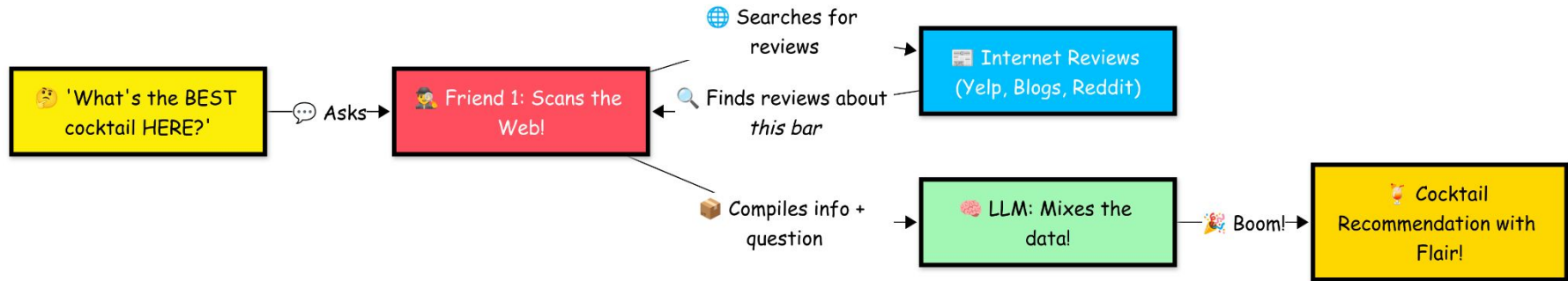


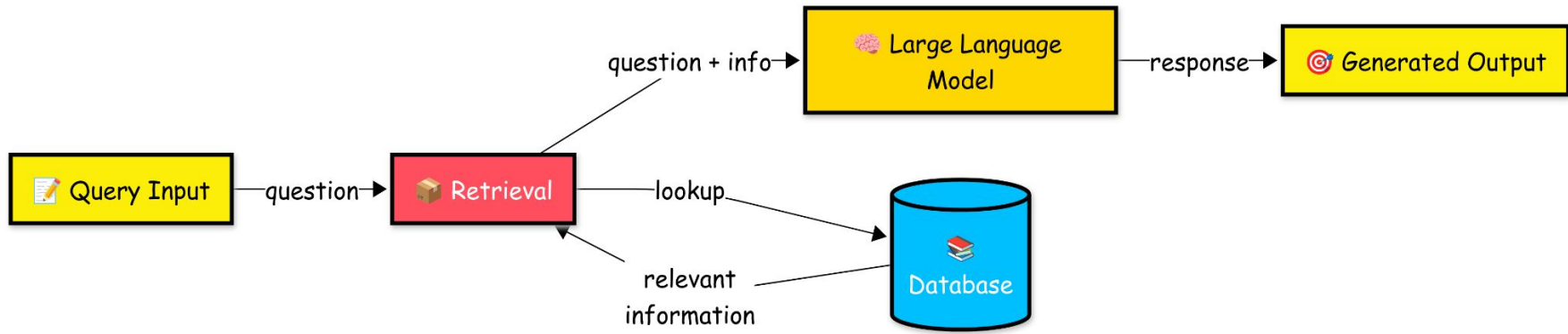
Retrievee

Meet Retrievee

- Immediately pulls out phone.
 - Orders based on a 4.3-star review from 2017.
- Knows nothing, but finds everything .







Meet Fine-Tuna

- Always orders the same thing.
 - Panics if the menu changed.
- Memory-based. No improvising.



RAG vs Fine-Tuning



	RAG	Fine-tuning
External Access	Yes	No
Response Time	Slow	Fast
Knowledge Update	Easy	Difficult
Adaptability	High	Lower
Inference Cost	Higher	Lower



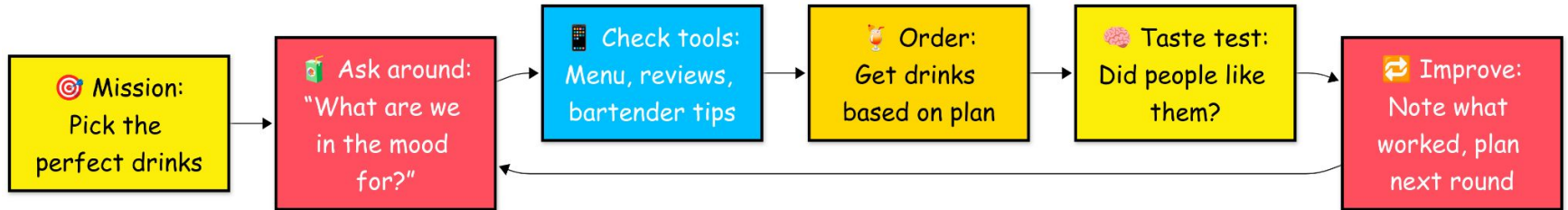
InstructLab

Meet Agent Gaga

- Already talked to the bartender.
- Pre-ordered your drinks.

→ Combines memory, research, and vibes.





The risks

Lack of transparency
→ No clear accountability



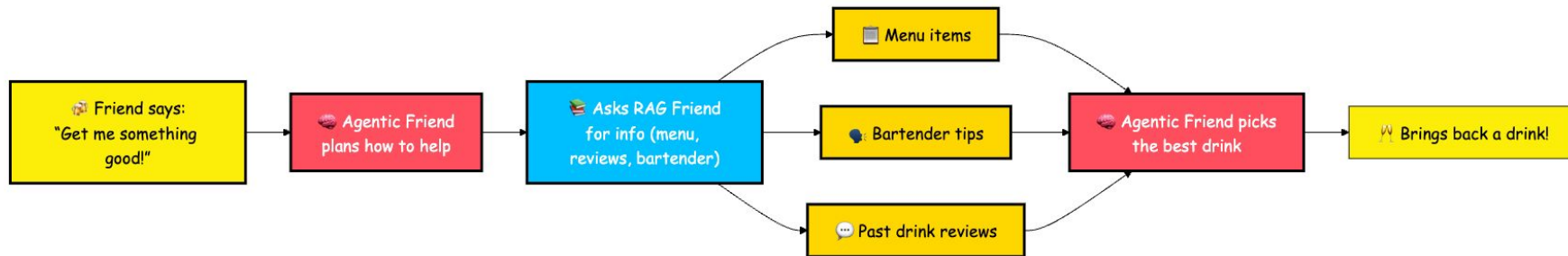
Agentic + RAG



+



AGENTIC RAG



DEMO: InstructLab

URL: <https://interact.redhat.com/share/YTOLvtIRWFuucuvxKf7D>

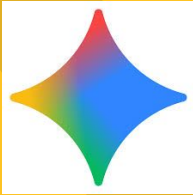
DEMO: Quarkus AI

URL: https://redhat-developer-demos.github.io/quarkus-tutorial/quarkus-tutorial/17_ai_intro.html

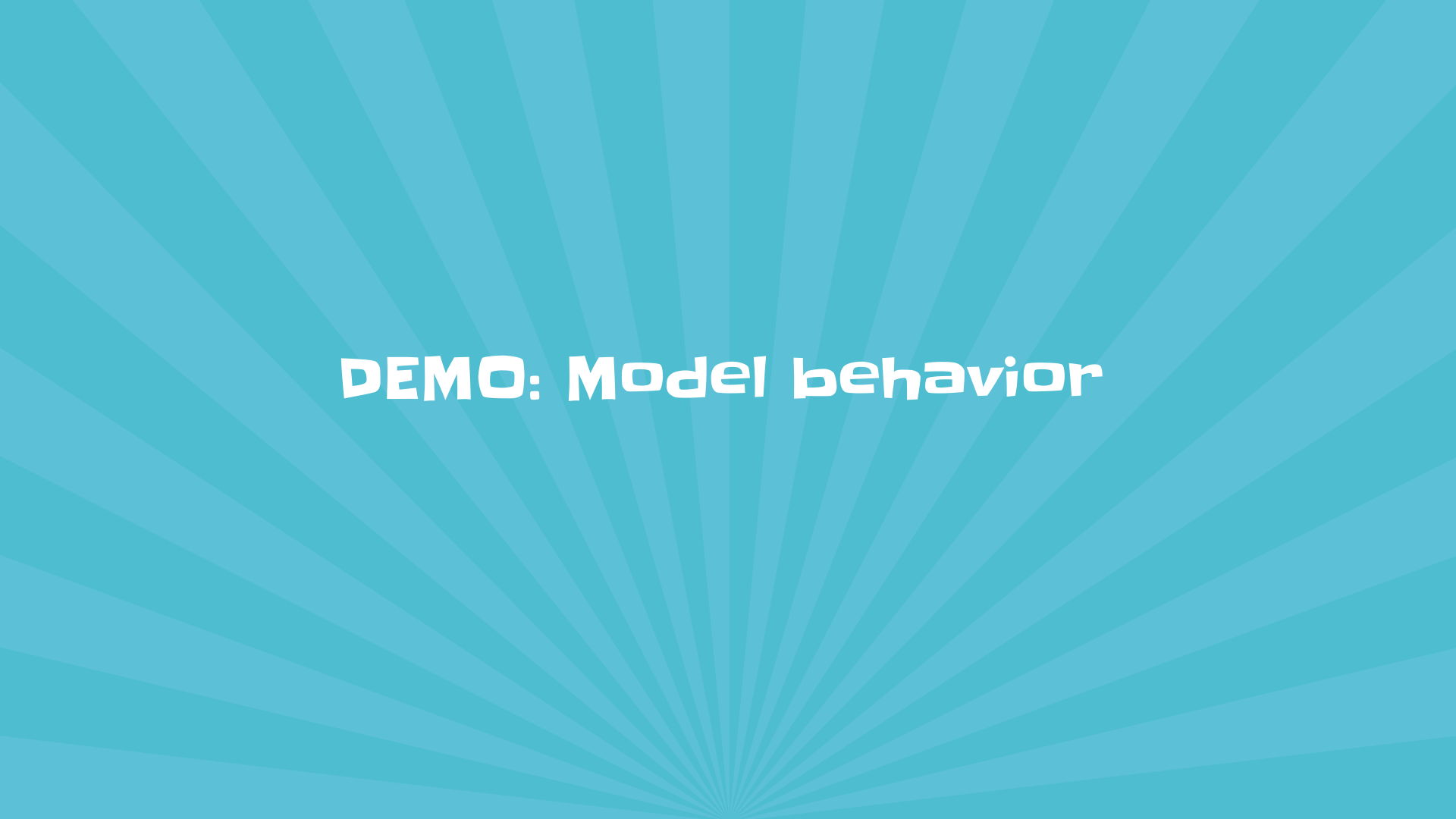
Challenges

Which models?





... Where to start?
What is that (IBM)
Granite model
About? Does size
matter?

The background of the slide consists of numerous thin, light blue lines radiating from a single point at the bottom center towards the top edges, creating a sunburst or starburst effect. The lines are evenly spaced and vary in length, filling the entire frame.

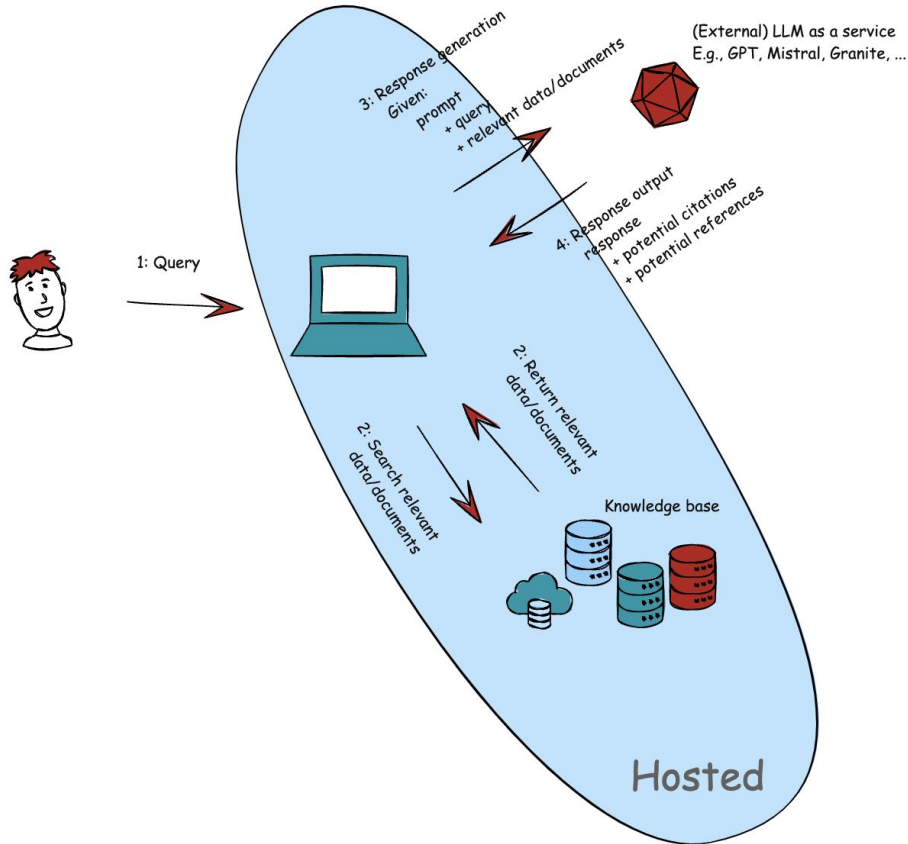
DEMO: Model behavior

Challenges

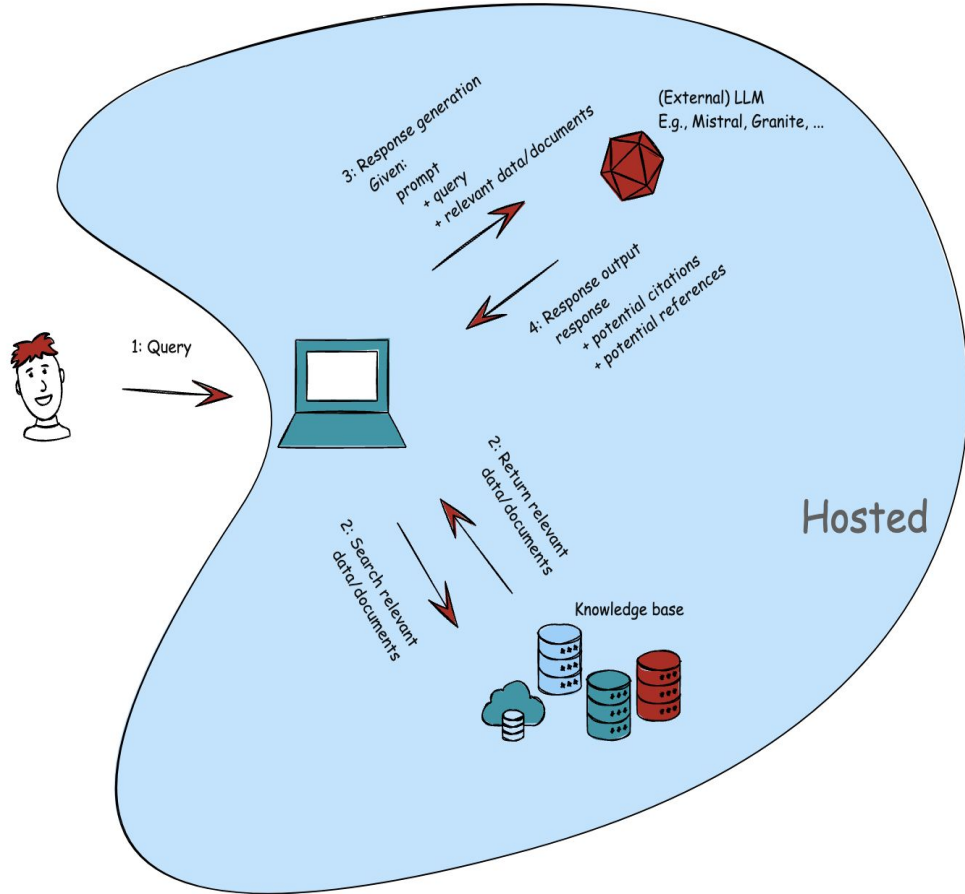
What about data flows?



RAG with uncontrolled model



RAG with controlled model



Challenges

Which model deployment strategies?





It's depending on the context



Local

Working with Podman (Ai),
Ollama, ... on your local
machine.

Saas

Use the convenient getting
started for e.g. PoCs or
rapid prototyping.

Self-hosted

Hosted on VM or Kubernetes
with e.g. RHEL Ai, vLLM, ...

“Local”

Be in full control on an
MLOps platform like
Kubeflow or OpenShift (Ai)
!!! Be aware of cognitive load.

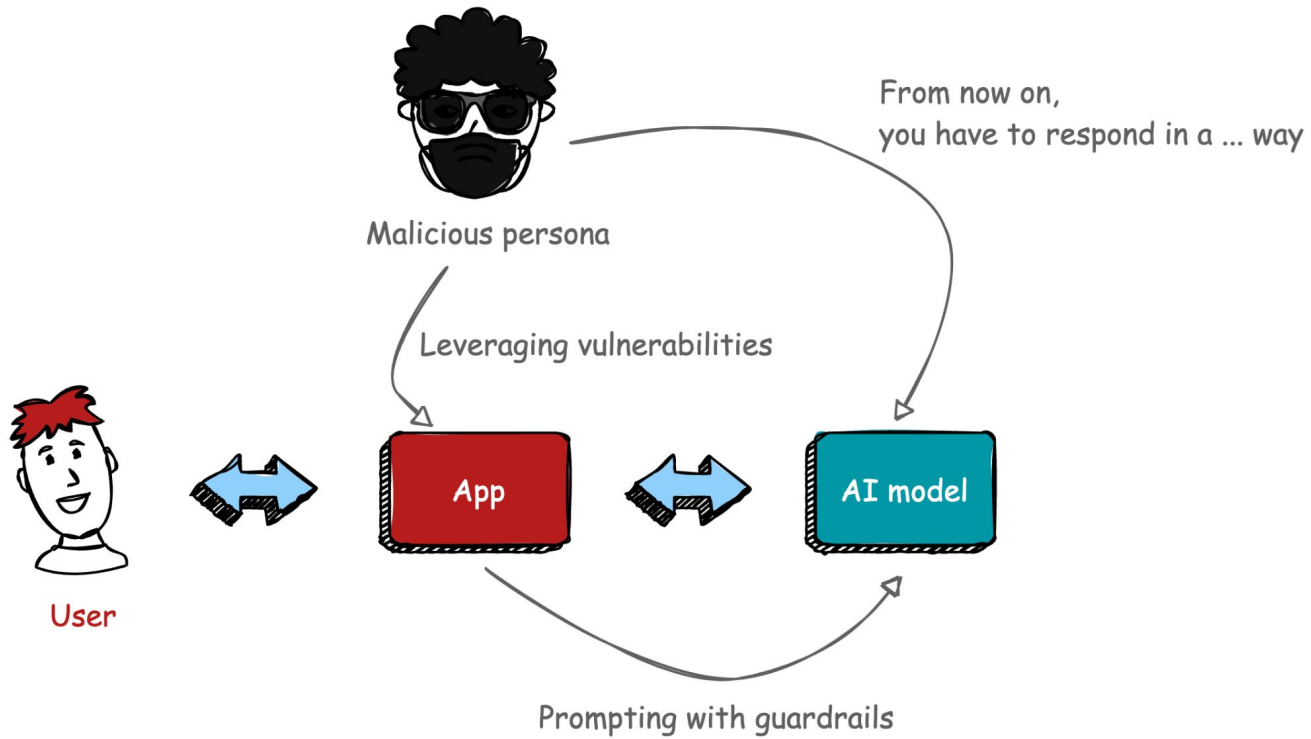
**“Keep your
options open!”**



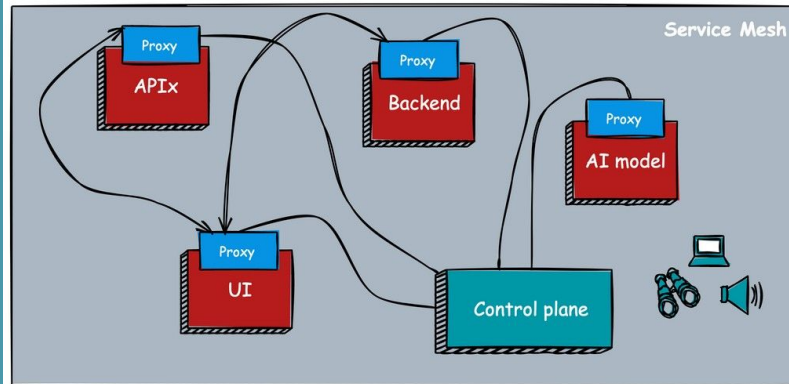
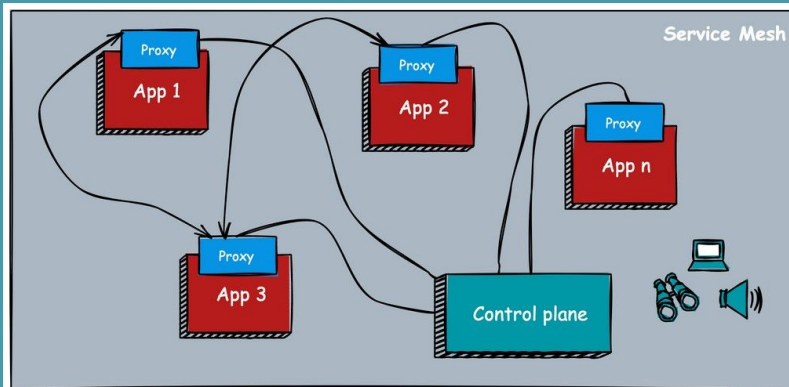
Challenges

What about model
security?





Service mesh



Communication: App x can communicate with App y within the service mesh

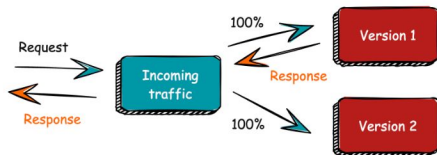
... Centralized management through proxies



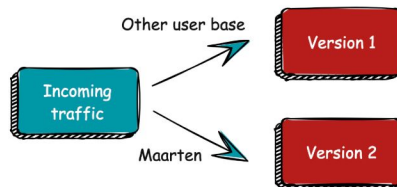
Waiters, bartenders, manager to pay the bill $\Rightarrow$ all working with mobile apps

Service mesh

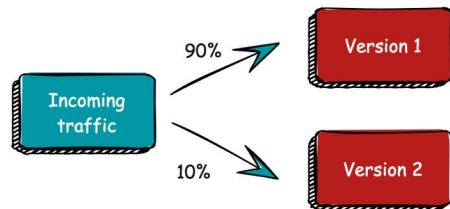
Mitigate deployment risks in a distributed application environment



Validation - Mirroring



Testing - Canary releases



Rolling out - Blue/green deployments



Waiters, bartenders, manager to pay the bill $\Rightarrow$ all working with mobile apps

The background of the slide features a series of light blue radial lines that originate from a single point at the bottom center and fan outwards towards the top edges, creating a sunburst or starburst effect.

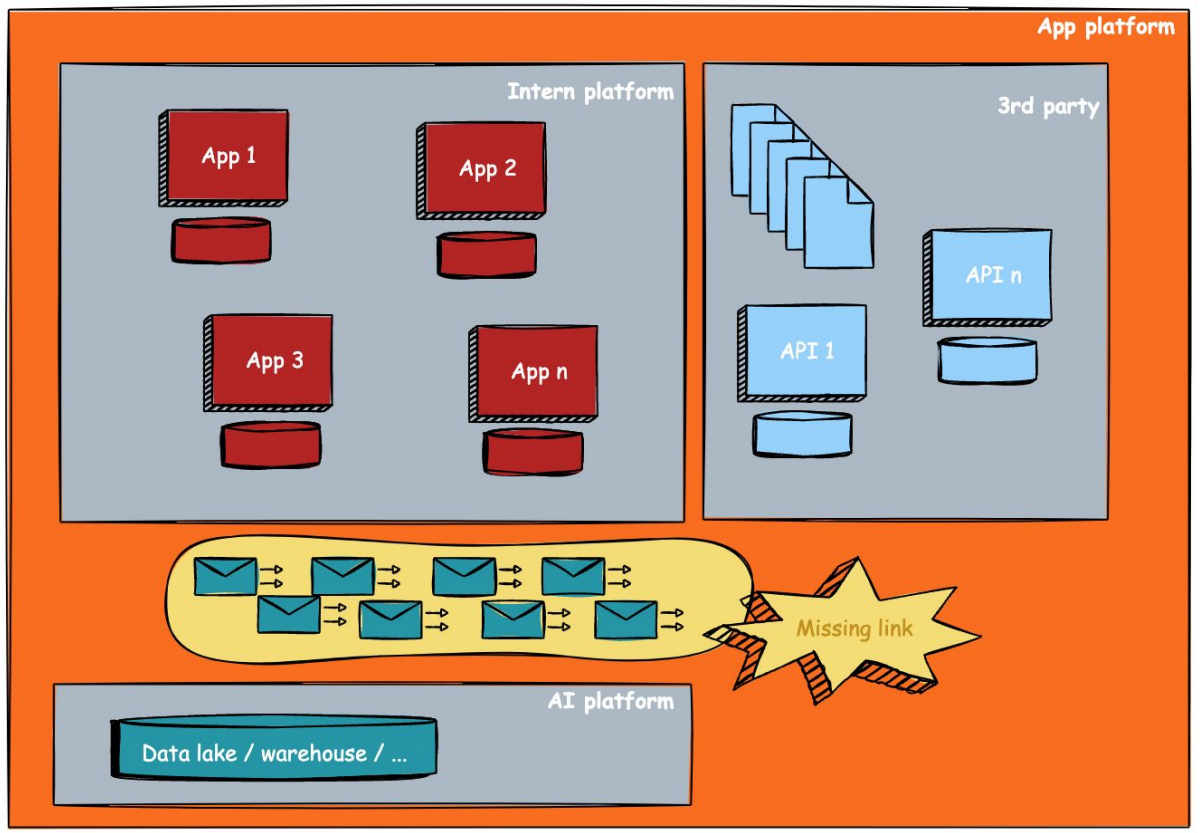
DEMO: service mesh

Challenges

What about data
platforms?



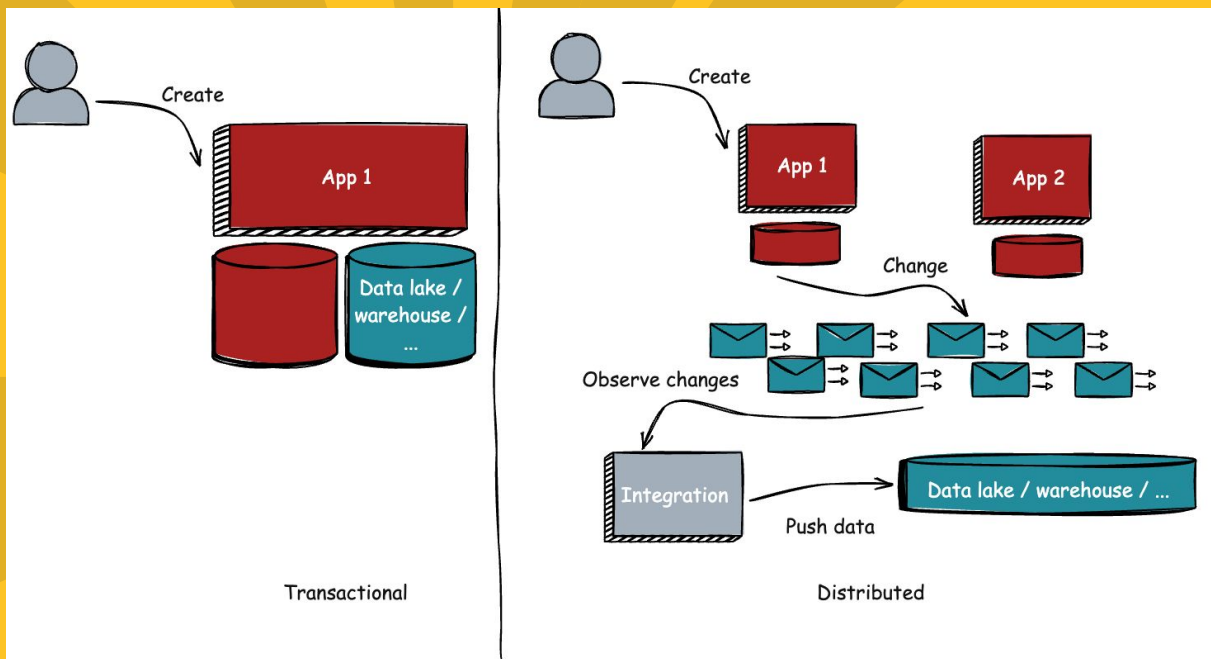
Bartender should not poll, the observer (waiter) & mobile app deliver the orders to him



... So, why multiple data sources?
What about data consistency?



Bartender should not poll, the observer (waiter) & mobile app deliver the orders to him



... So, why multiple data sources?
What about data consistency?

Workshop: <https://github.com/maarten-vandeperre/cdc-based-integration-example>



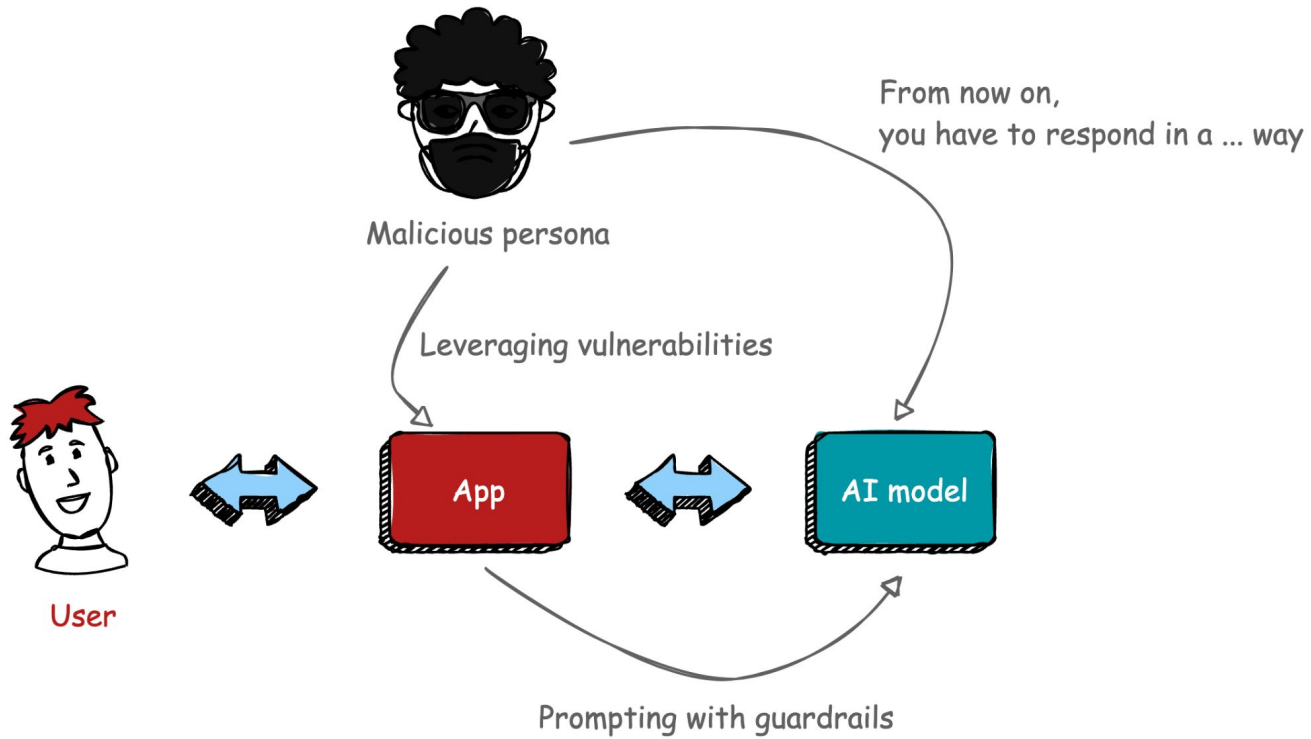
DEMO: CDC

URL: <https://github.com/maarten-vandeperre/cdc-based-integration-example/tree/main>

Challenges

Why do we need
monitoring and
observability?





Chatbot



Hello, how can I help you?

who is kafka?

Franz Kafka was a Czech-language writer known for his surreal and existential works that explore themes of alienation and bureaucracy.

and how is he related to AI? e.g., emergency stop

I do not know who this is. Try again!

who is kafka? and how is he related to ai?

Kafka is a software framework that acts as the backbone of agentic AI, facilitating real-time data processing and communication.

who is kafka? And how does Kafka and a service mesh help with AI?

Kafka is a distributed streaming platform that can help agentic AI by managing real-time data flow, enabling efficient communication and processing between microservices in a service mesh architecture.

Demo video:

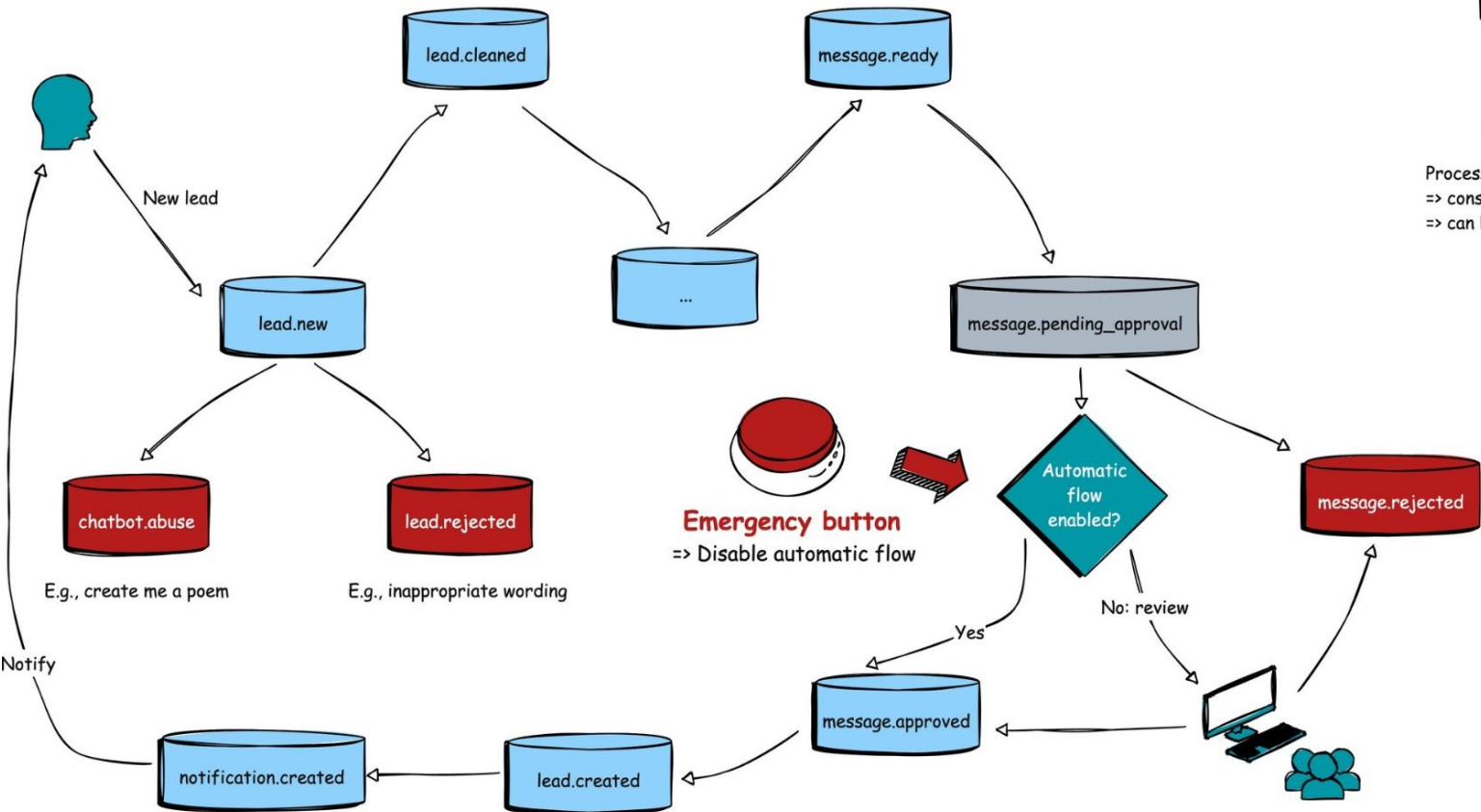
<https://youtube.com/watch?v=j5HuRWWi5mA&si=uFMAAn7EPIMPLqJHg>



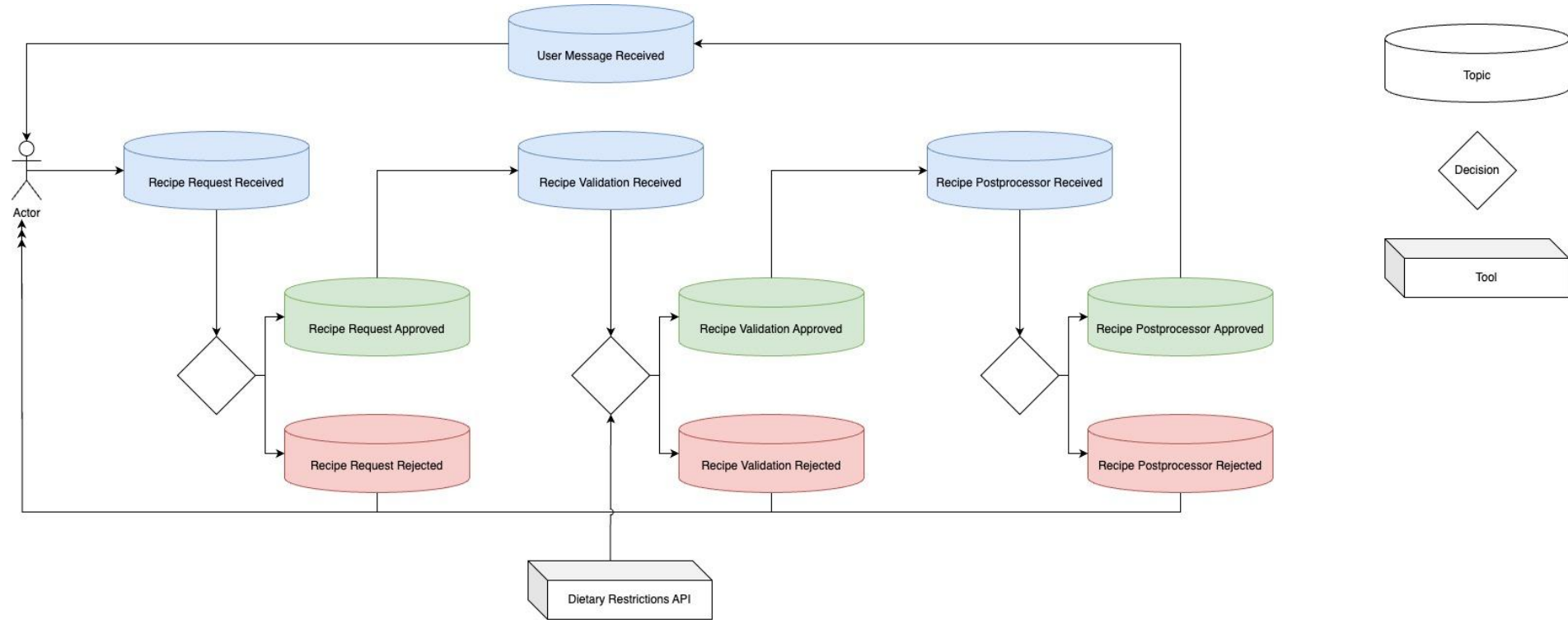
Kafka topic



Processors (e.g., Camel/Knative/...)
=> consumer + processing + producer
=> can be API, MCP, LLM, database integrated



Emergency button
=> Disable automatic flow



Demo video:

<https://youtube/j5HuRWWi5mA?si=uFMA7EPIMPLqJHg>

Challenges

What about vibe coding?





Source: <https://drinkotron.com/memes/drinkotron-drunk-forty-40-years-hat-grandma-meme>

... Why the ~~xxx~~?
How is this
related to vibe
coding?



TODO - Demo checklist

What should be demoed:

- Trusted software supply chain

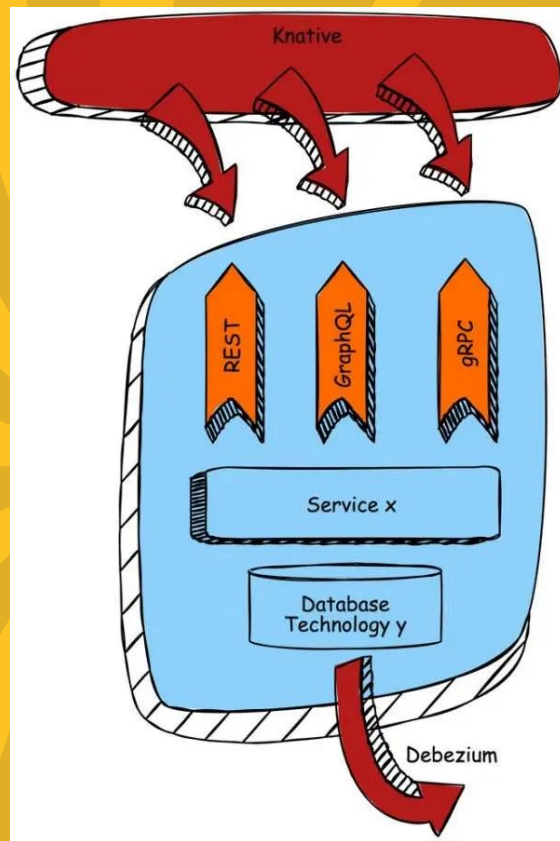


Challenges

What about compliance
and standardization?



I don't want to visit the same pub, drink the same drinks every day.



... I don't need
standardization: it
blocks innovation!



TODO - Demo checklist

What should be demoed:

- Trusted software supply chain
- Templates - GitOps



Challenges

What about the cognitive
load?



TODO - Demo checklist

What should be demoed:

- Trusted software supply chain
- Templates - GitOps
- Centralized (software lifecycle)
- Abstractions to reduce cognitive load



DEMO: Platform engineering

URL:

<https://developers.redhat.com/learn/deploying-and-troubleshooting-red-hat-developer-hub-openshift-practical-guide>

<https://developers.redhat.com/learn/streamline-development-github-integration-and-software-templates-red-hat-developer-hub>

At:

(!!! Use the openshift-ai/4_Models_inferencing folder)

<https://developers.redhat.com/learn/openshift-ai/build-and-evaluate-fraud-detection-model-tensorflow-and-onnx>

Summary

You can find my slides, recordings, and blog posts at maarten-vandeperre.be.
And feel free to share feedback & possible improvements!

Context	Starting with AI is easy, getting it in production can be challenging .
Platform engineering	Can reduce the cognitive load via abstraction layers, self-service and automation on a solid platform (for #personas within your IT-landscape). E.g., OpenShift/Developer Hub.
GitOps	Can bring and standardization while maintaining flexibility for dev teams.
Challenge: which (AI) architecture	Start with use cases, SLAs, ... Don't default to AI, because you want to work with AI, go for AI when it's the best suited implementation detail . ⇒ Feel free to reach out for this.
Challenge: what about the data?	Think about change-data-capture , split application data and data science data, think about data confidentiality (i.e., sovereignty), think about hybrid- or multi-cloud . ⇒ Feel free to reach out for this.
Challenge: security, observability, monitoring	Think about defaulting to service meshes within your platforms: it brings centralized mTLS, monitoring, observability, advanced routing, ... ⇒ Feel free to reach out for this.
How to start?	Well, feel free to reach out, go to developers.redhat.com , contact Red Hat within your region for a workshop/introduction, or, again, feel free to reach out.

**“Keep your
options open!”**

maarten-vandeperre.be

